

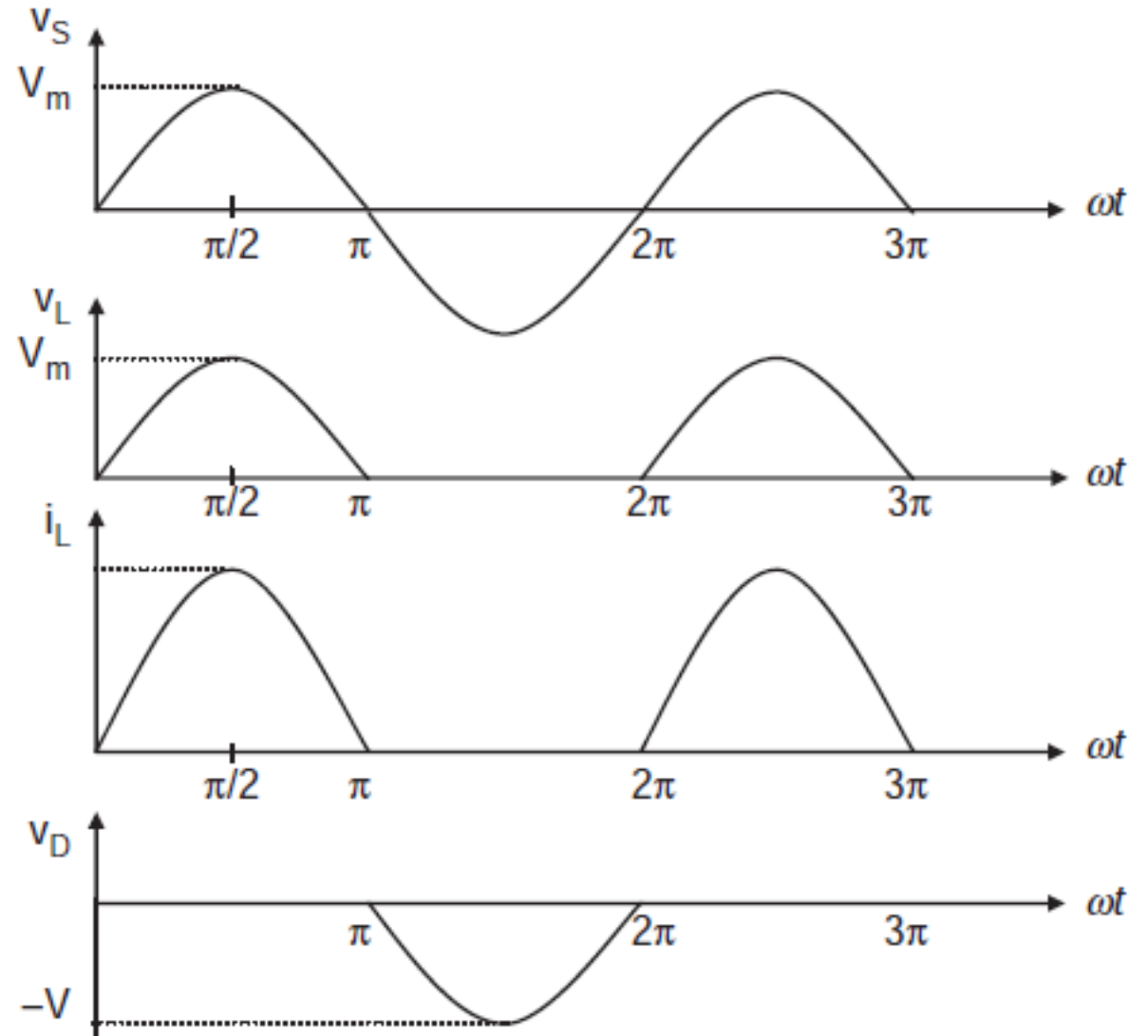
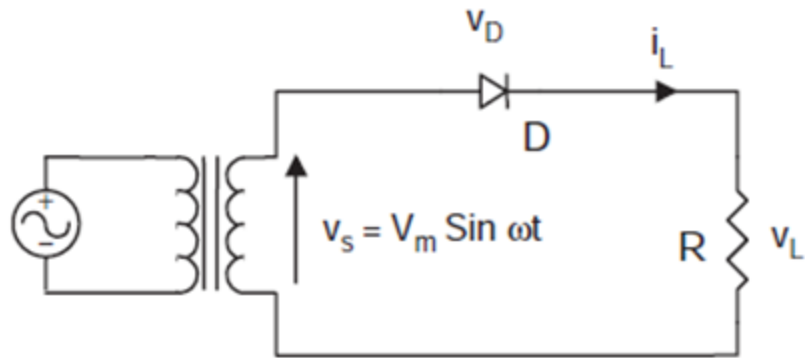
POWER ELECTRONICS

Uncontrolled
Rectifiers

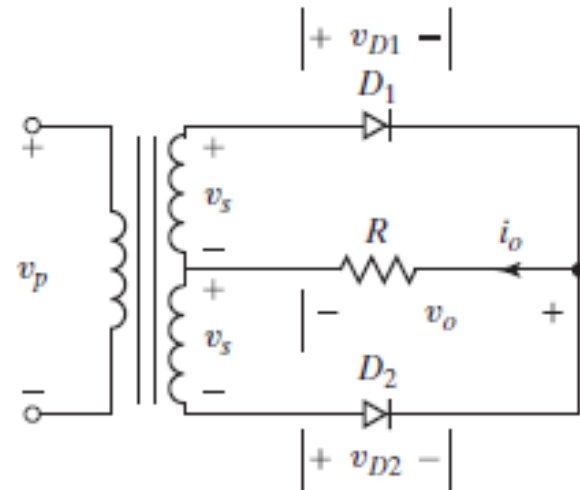
Single Phase Uncontrolled Rectifiers

- Single-Phase Half-Wave Rectifier
- Single-Phase Full-Wave Rectifier

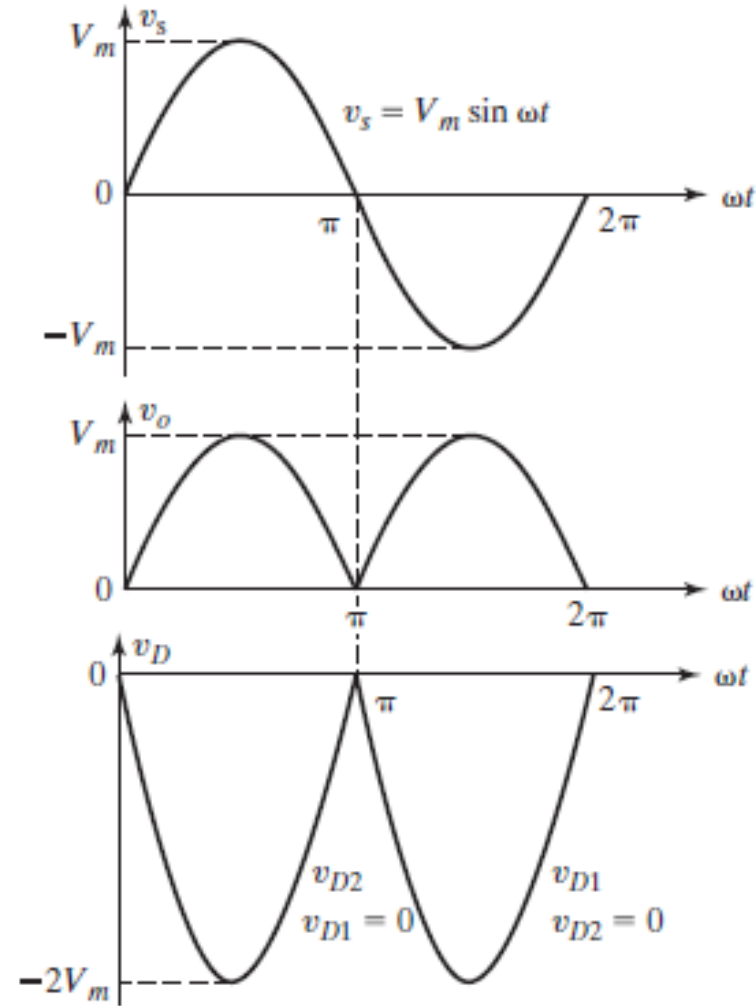
Single-Phase Half-Wave Rectifier



Single-Phase Full-Wave Rectifier



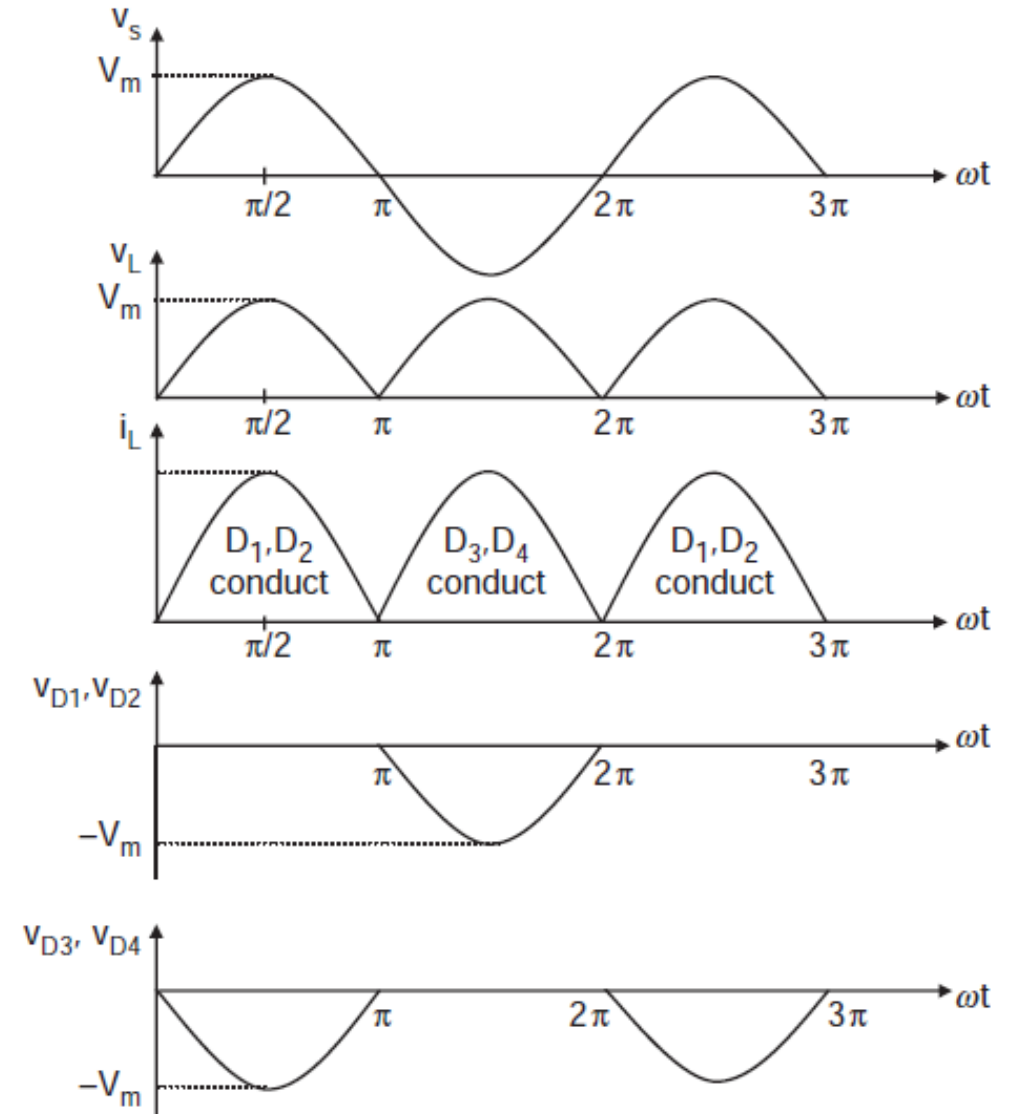
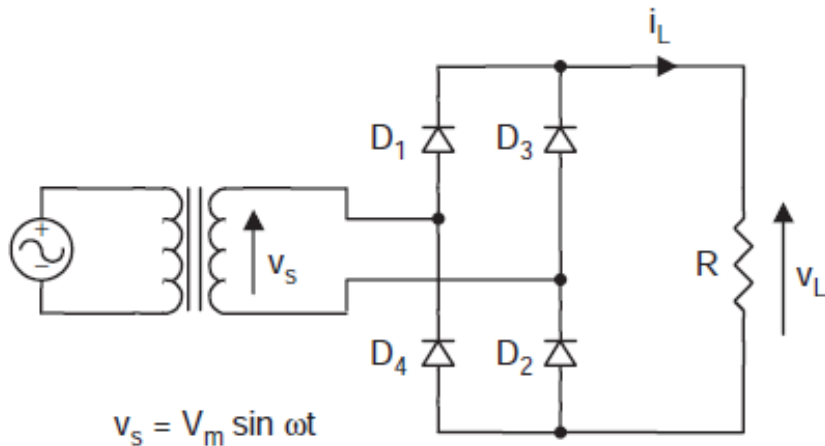
(a) Circuit diagram



(b) Waveforms

Single-Phase Bridge Rectifier

- No center-tapped transformer
- PIV rating reduced from $2V_m$ to V_m



Voltage Relationships

- Average value of output voltage is V_{dc} . How it is defined?
 - For half wave?
 - For full wave?
- Root Mean Square value of output voltage is V_{rms} . How it is defined?
 - For half wave?
 - For full wave?

Current Relationships

- Average value of load current is I_{dc} and since the load is purely resistive. It can be defined as: ?
 - For half wave?
 - For full wave?
- Root Mean Square value of load current is I_{rms} . How it is defined?
 - For half wave?
 - For full wave?

Power Relationships

- Output dc power is $P_{dc} = V_{dc}I_{dc}$
- Output ac power is $P_{ac} = V_{rms}I_{rms}$

Rectification Ratio (Conversion Efficiency)

- Figure of merit for comparing the effectiveness of rectification. It is defined as:

$$\sigma = \frac{P_{dc}}{P_{ac}} = \frac{V_{dc}I_{dc}}{V_{rms}I_{rms}}$$

- For half wave?
- For full wave?

Form Factor

- Ratio of the root mean square value of a voltage or current to its average value. It is defined as:

$$FF = \frac{V_{rms}}{V_{dc}}$$

- For half wave?
- For full wave?

Ripple Factor

- It is the measure of ripple content. It is defined as:

$$RF = \frac{V_{ac}}{V_{dc}}$$

- Where V_{ac} is the effective (rms) value of the ac component of load voltage v_L

$$V_{ac} = \sqrt{V_{rms}^2 - V_{dc}^2}$$

- Substituting into the equation of RF,

$$RF = \sqrt{\left(\frac{V_{rms}}{V_{dc}}\right)^2 - 1} = \sqrt{FF^2 - 1}$$

- For half wave?
- For full wave?

Transformer Utilization Factor

- TUF is the measure of merit of rectifier circuit. It is defined as the ratio of the dc output power to the transformer volt-amp rating required by the secondary winding. Mathematically:

$$TUF = \frac{P_{dc}}{V_s I_s} = \frac{V_{dc} I_{dc}}{V_s I_s}$$

- Where V_s and I_s are the rms voltage and rms current ratings of the transformer secondary.

$$V_s = \frac{V_m}{\sqrt{2}} = 0.707V_m$$

- I_s is the same as that of the load current I_L

- For half wave?
- For full wave?
- For bridge rectifier?

Power Factor

- Power factor is related as:

$$PF = \frac{P_{ac}}{V_s I_s}$$

Crest Factor

- Crest factor is measure of the peak input current $I_{s(peak)}$ as compared with its rms value I_s

$$CF = \frac{I_{s(peak)}}{I_s}$$

Design Considerations

1. Represent rms input voltage of the rectifier in terms of output average voltage V_{dc}
 - For half wave?
 - For full wave?
2. Peak repetitive reverse voltage V_{RRM} in terms of output average voltage V_{dc}
 - For half wave?
 - For full wave?
 - For bridge rectifier?
3. Peak repetitive forward current I_{FRM} in terms of output average current I_{dc}
 - For half wave?
 - For full wave?

Design Considerations

Important design parameters of basic single-phase rectifier circuits with resistive load

	Half-wave rectifier	Full-wave rectifier with center-tapped transformer	Full-wave bridge rectifier
Peak repetitive reverse voltage V_{RRM}	$3.14V_{dc}$	$3.14V_{dc}$	$1.57V_{dc}$
RMS input voltage per transformer leg V_s	$2.22V_{dc}$	$1.11V_{dc}$	$1.11V_{dc}$
Diode average current $I_{F(AV)}$	$1.00I_{dc}$	$0.50I_{dc}$	$0.50I_{dc}$
Peak repetitive forward current I_{FRM}	$3.14I_{F(AV)}$	$1.57I_{F(AV)}$	$1.57I_{F(AV)}$
Diode rms current $I_{F(RMS)}$	$1.57I_{dc}$	$0.785I_{dc}$	$0.785I_{dc}$
Form factor of diode current $I_{F(RMS)}/I_{F(AV)}$	1.57	1.57	1.57
Rectification ratio	0.405	0.81	0.81
Form factor	1.57	1.11	1.11
Ripple factor	1.21	0.482	0.482
Transformer rating primary VA	$2.69P_{dc}$	$1.23P_{dc}$	$1.23P_{dc}$
Transformer rating secondary VA	$3.49P_{dc}$	$1.75P_{dc}$	$1.23P_{dc}$

Example 1

If a full-wave rectifier with center-tapped transformer has purely resistive load, determine (a) efficiency, (b) FF, (c) RF, (d) TUF, (e) PIV of diode D_1 , (f) CF of the input current, and (g) input power factor PF

Single- Φ Full Wave Rectifier With RL Load

- With inductive load, the input to the rectifier will no longer a sine wave
- If Φ is the angle between the fundamental components of the input current and voltage, Φ is called the displacement angle.

- The displacement factor is defined as:

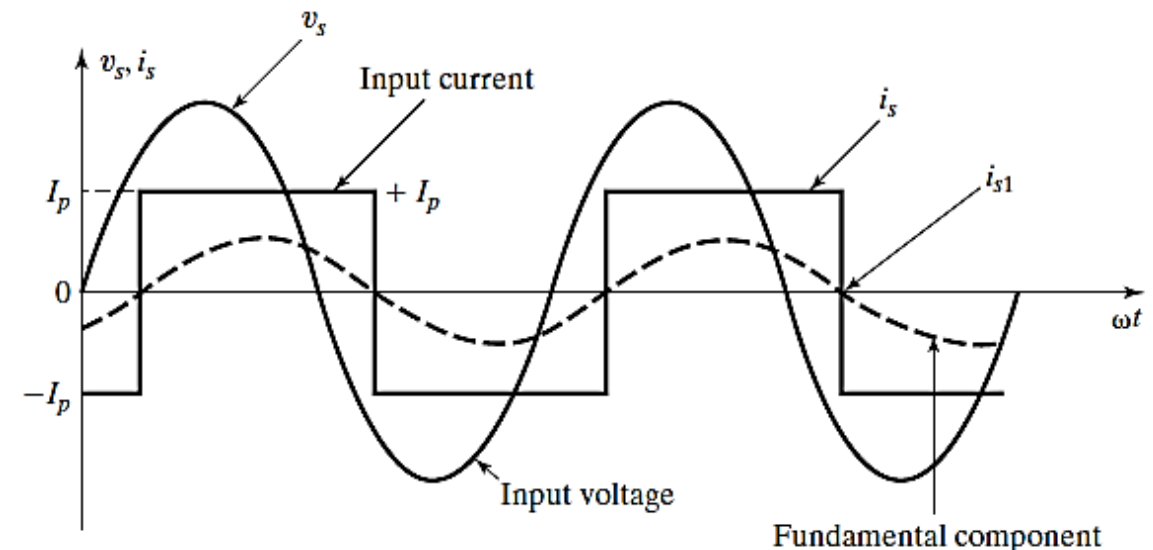
- $DF = \cos \Phi$

- The harmonic factor (HF) of the input current is defined as:

- $HF = \left(\frac{I_S^2 - I_{S1}^2}{I_{S1}^2} \right)^{1/2} = \left[\left(\frac{I_S}{I_{S1}} \right)^2 - 1 \right]^{1/2}$

- Where I_{S1} is the fundamental component of the input current I_S . The input PF is defined as:

- $PF = \frac{I_{S1}}{I_S} \cos \Phi$

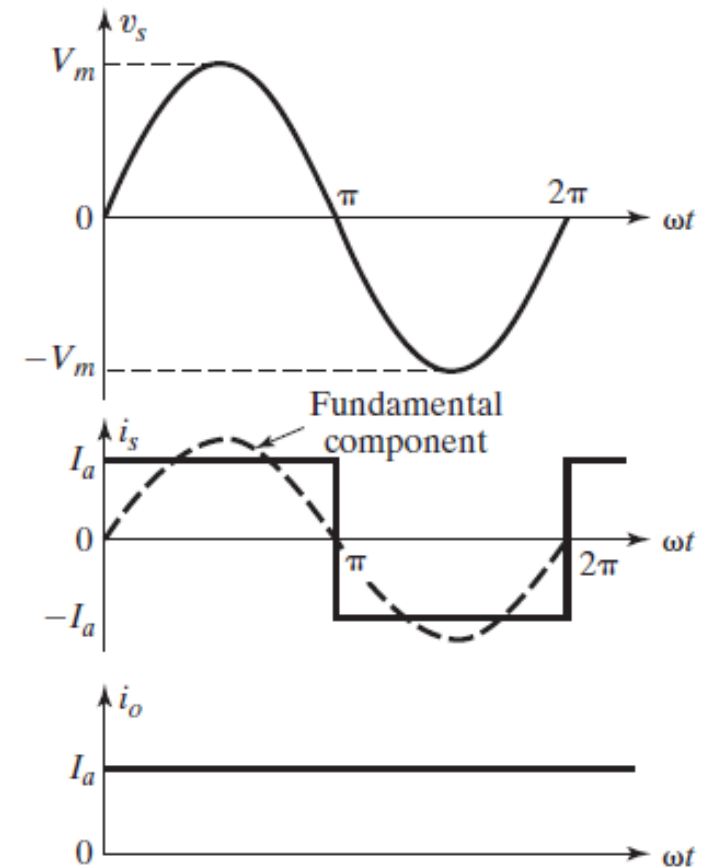
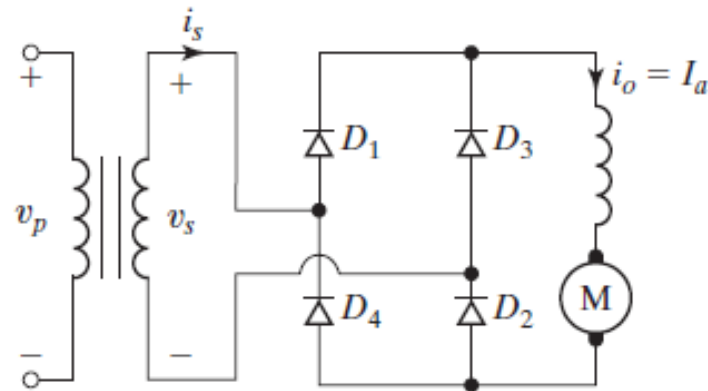


Single- Φ Full Wave Rectifier With RL Load

- HF is the measure of distortion of a waveform and is also known as total harmonic distortion (THD)
- If the input current is purely sinusoidal, the power factor equals the displacement factor DF and the displacement angle becomes the impedance angle for RL load
- Displacement factor is also known as displacement power factor (DPF)
- An ideal rectifier should have $\sigma = 100\%$, $V_{ac} = 0$, $RF = 0$, $TUF = 1$, $HF = THD = 0$, and $PF = DPF = 1$

Example 2

- A single-phase bridge rectifier that supplies a very high inductive load such as a dc motor is shown in figure. The turns ratio of the transformer is unity. The load is such that the motor draws a ripple-free armature current. Determine the HF of the input current and the input PF of the rectifier. The fundamental component of the input current $I_{s1} = 0.9I_a$

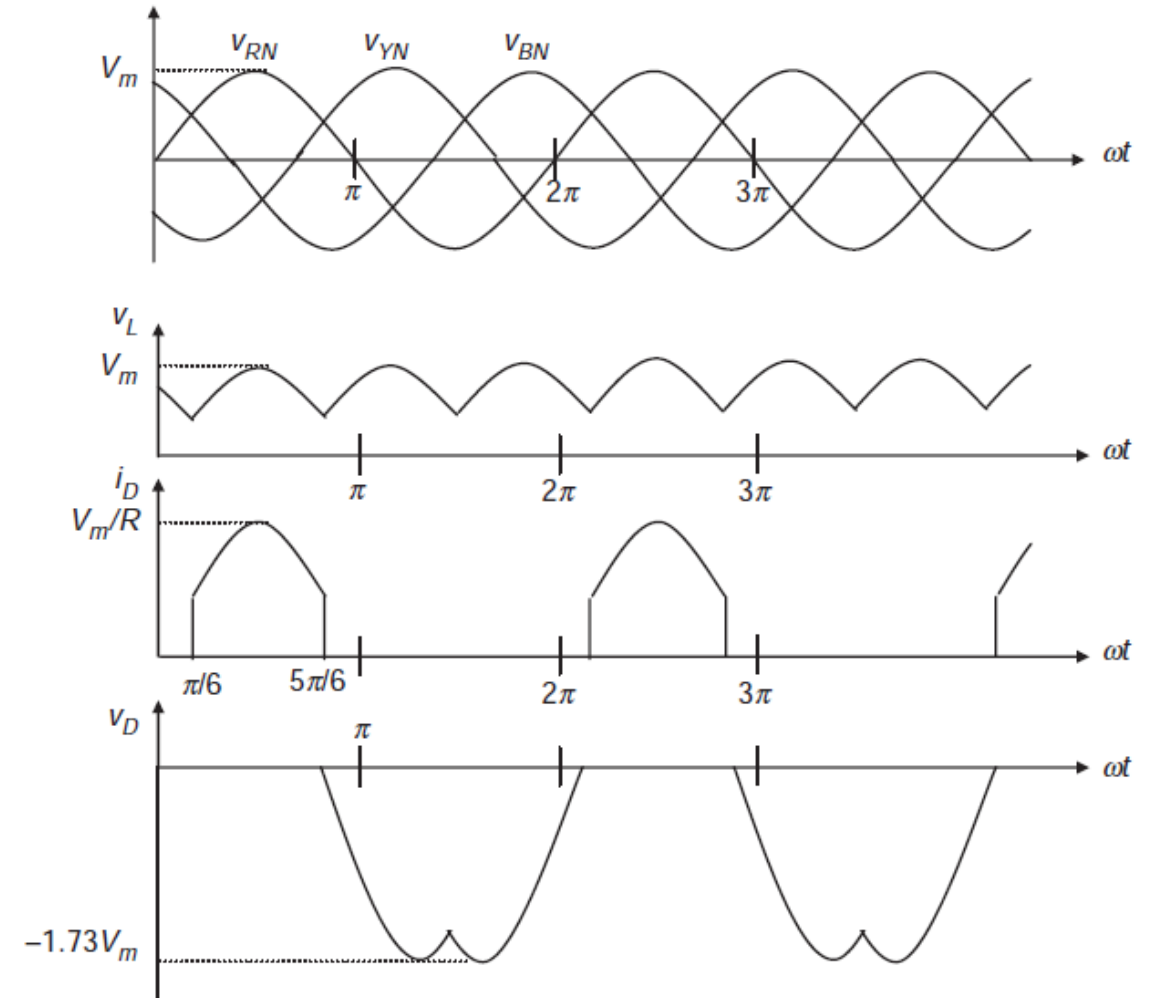
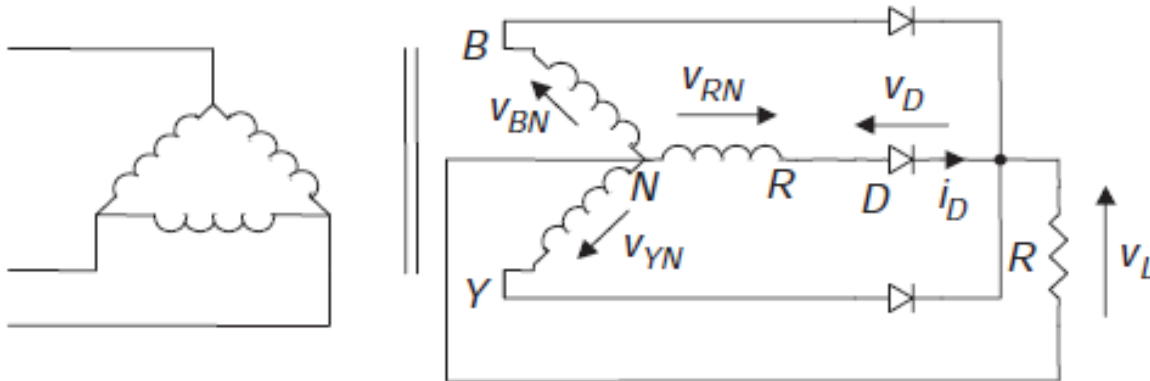


Three-Phase Uncontrolled Rectifiers

- Single-phase rectifiers are used only for low to medium power applications
- For output higher than 15 kW, three-phase rectifiers are used
- Types:
 - Star rectifiers
 - Bridge rectifiers

Three-Phase Star Rectifiers (Half-Wave)

- Similar to three single-phase half-wave rectifiers combined together
- Diode is on when:
 - Voltage on that phase is greater than the other two
- Diode conduction angle is $2\pi/3$ instead of π
- Average value of output voltage?
- RMS value of output voltage?
- rms current in each secondary winding?

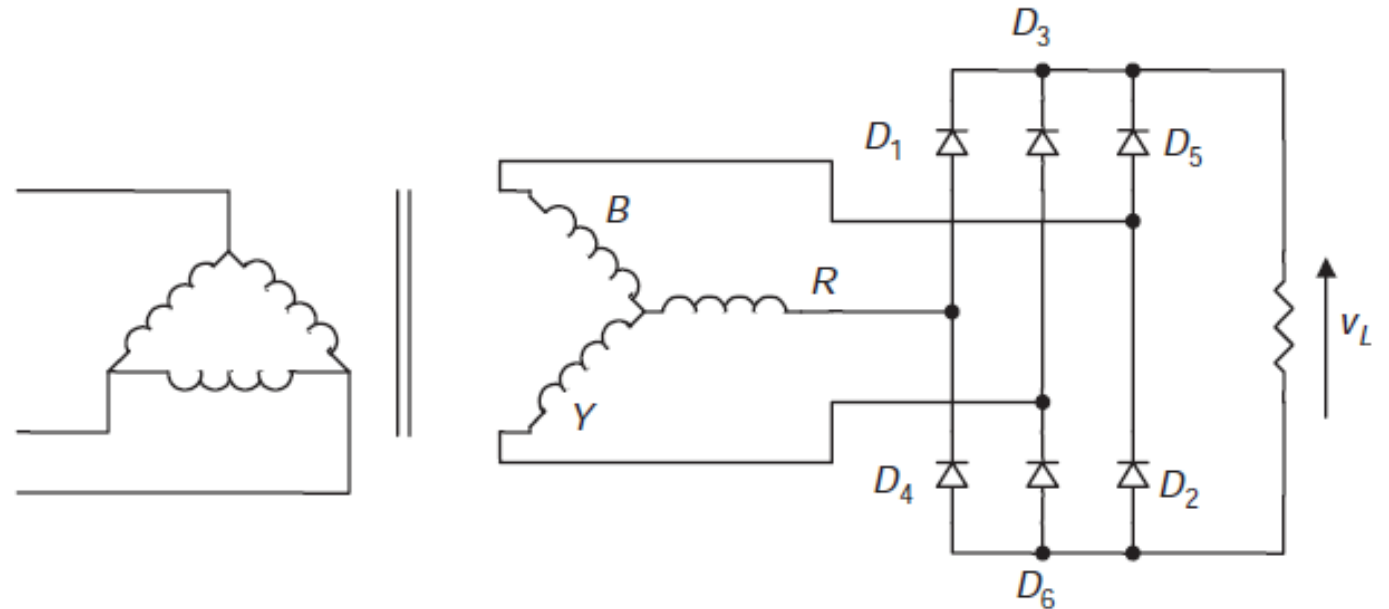


Example 3

- A three-phase star rectifier has a purely resistive load with R ohms. Determine (a) the efficiency, (b) the FF, (c) the RF, (d) the TUF, (e) the PIV of each diode, and (f) the peak current through a diode if the rectifier delivers $I_{dc} = 30$ A at an output voltage of $V_{dc} = 140$ V

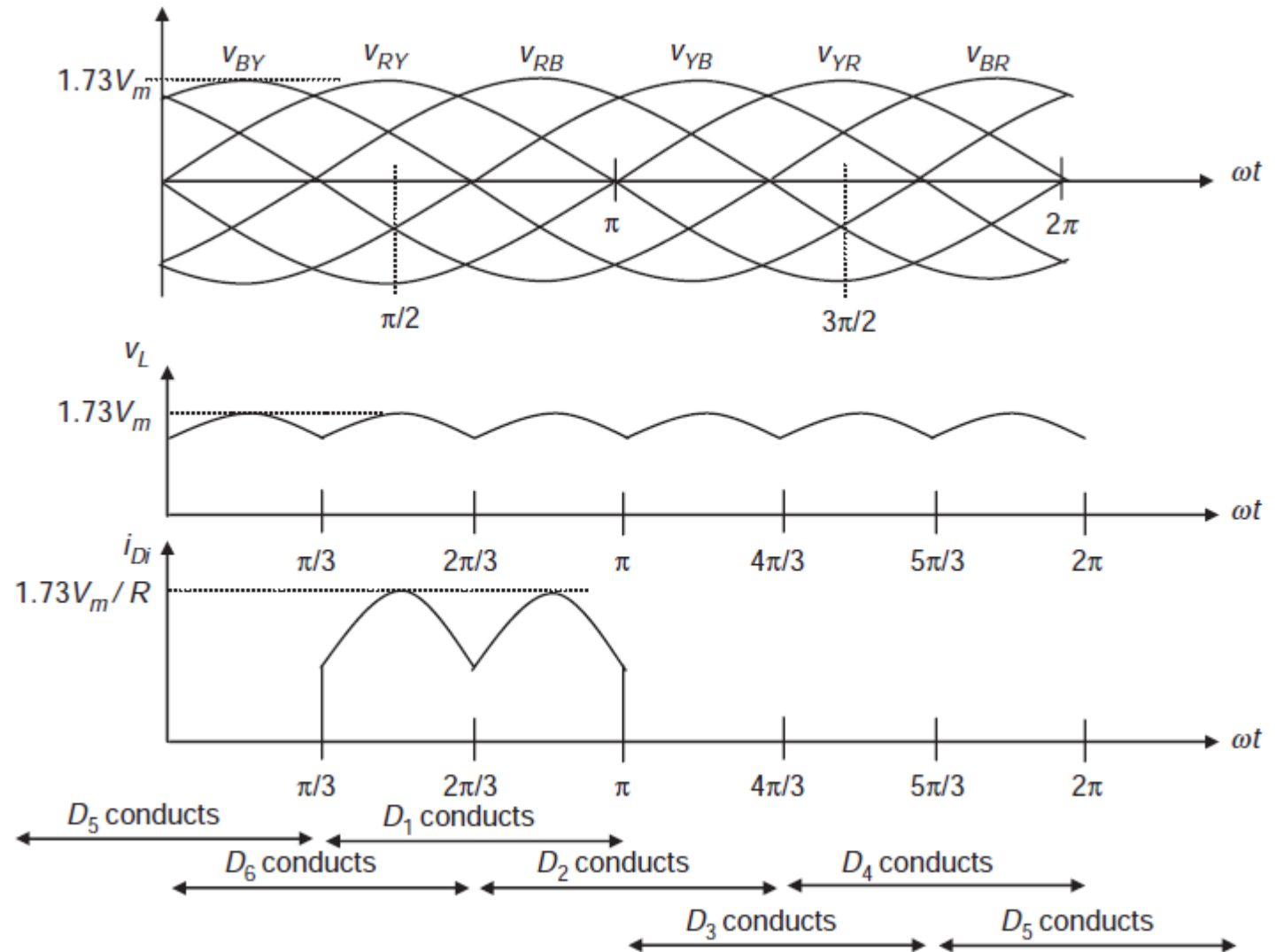
Three-Phase Bridge Rectifiers

- Commonly used for high power applications
- Have highest possible TUF
- Diodes are numbered in the order of conduction sequences
- Conduction angle is $2\pi/3$
- Line voltage and phase voltage concept



Three-Phase Bridge Rectifiers

- Average value of output voltage?
- RMS value of output voltage?
- Diode rms current?
- Each secondary rms current?

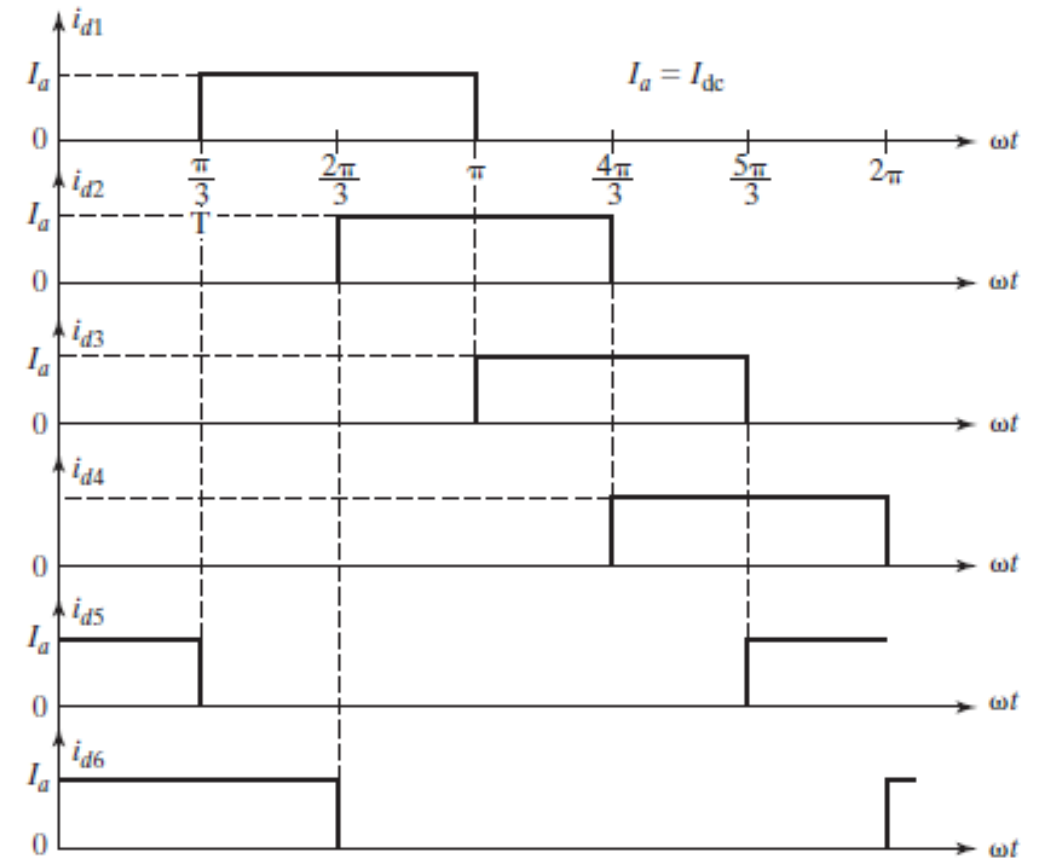


Example 4

- A three-phase bridge rectifier has a purely resistive load of R . Determine (a) the efficiency, (b) the FF, (c) the RF, (d) the TUF, (e) the peak inverse voltage of each diode, and (f) the peak current through a diode. The rectifier delivers $I_{dc} = 60$ A at an output voltage of $V_{dc} = 280.7$ V and the source frequency is 60 Hz.

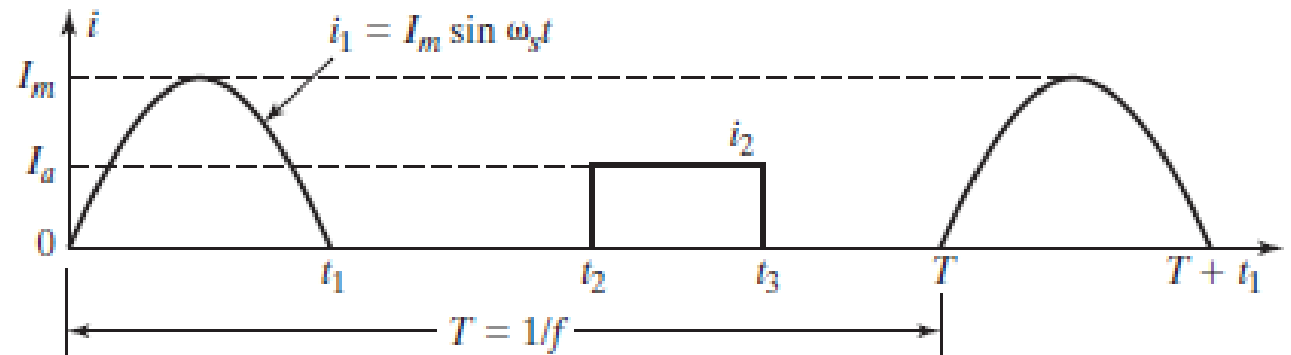
Example 5

- A three-phase bridge rectifier supplies a highly inductive load such that the average load current is $I_{dc} = 60$ A and the ripple content is negligible. Determine the ratings of the diodes if the line-to-neutral voltage of the Y-connected supply is 120 V at 60 Hz.



Example 6

- The current through a diode is shown in figure below. Determine (a) the rms current and (b) the average diode current if $t_1 = 100 \mu\text{s}$, $t_2 = 350 \mu\text{s}$, $t_3 = 500 \mu\text{s}$, $f = 250 \text{ Hz}$, $f_s = 5 \text{ kHz}$, $I_m = 450 \text{ A}$, and $I_a = 150 \text{ A}$.



Reference

Power Electronics Handbook
Haroon Rashid

Power Electronics
Ned Mohan